

Credit Risk Intelligence Platform AI-Powered Underwriting & Explainability

NeoStats AI Engineer Candidate Assignment

Prepared By:

Harsha
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Business Problem & Objectives

Banking Challenges

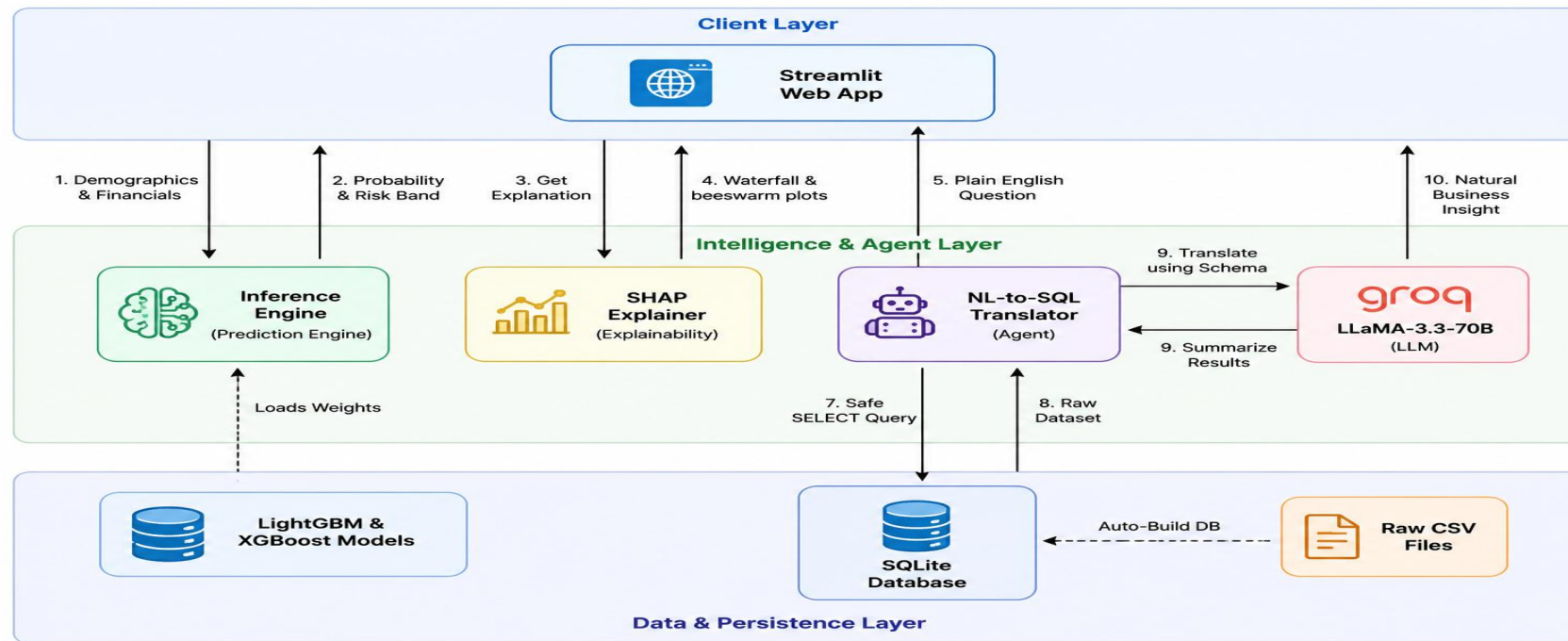
- Manual credit assessment is time-consuming and inconsistent.
- High-risk applicants can result in significant financial losses.
- Regulatory compliance requires transparent and explainable decisions.
- Business teams require quick access to actionable insights.

Project Objectives

- Predict loan default probability using machine learning.
- Provide explainable risk assessments through SHAP analysis.
- Enable natural-language data exploration using an AI-powered chatbot.
- Transform model predictions into business-readable underwriting rules.



System Architecture Overview



Key Components

- Streamlit
- LightGBM
- XGBoost
- SHAP
- Groq LLM

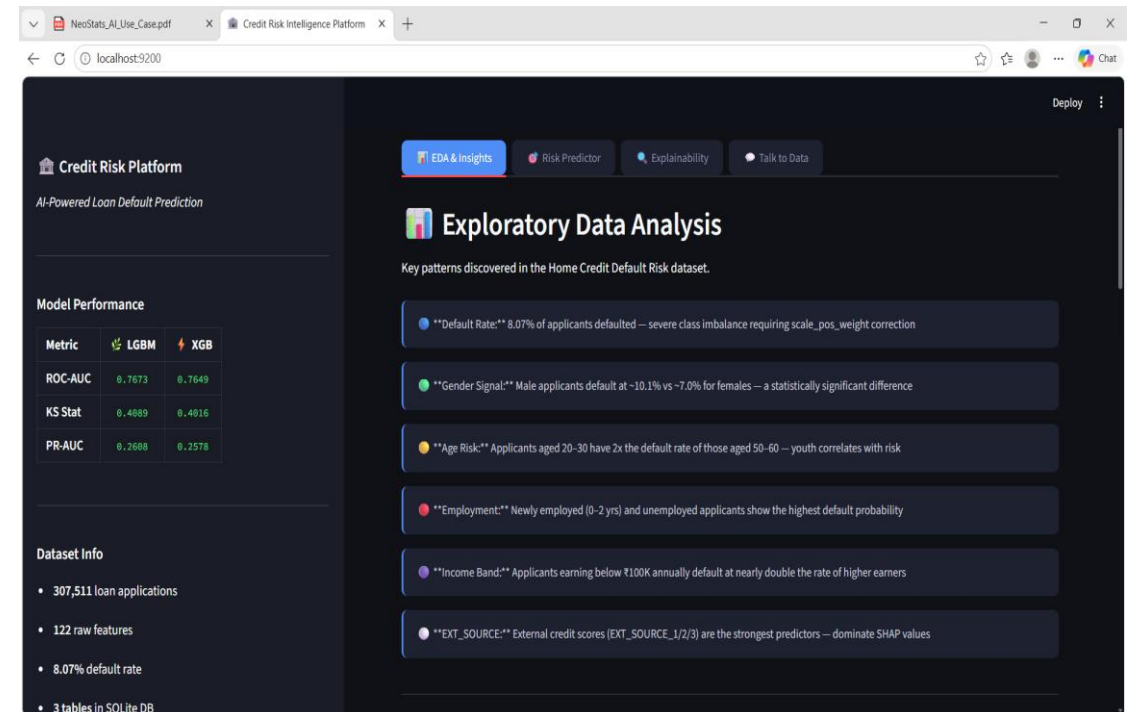
Data Understanding & Exploratory Data Analysis

Dataset Overview

- Home Credit Default Risk Dataset
- Over 307,000 loan applications
- Default Rate: 8.07%

Key Insights

- Significant class imbalance exists in the dataset.
- External credit scores are strong predictors of default.
- Income and employment history influence repayment behavior.
- Risk increases with higher debt-service obligations.



Feature Engineering & ML Pipeline

Engineered Features

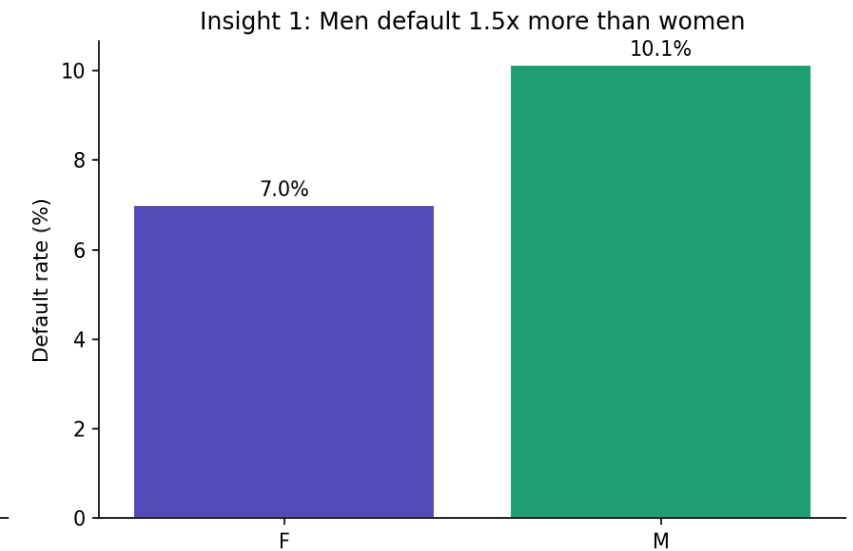
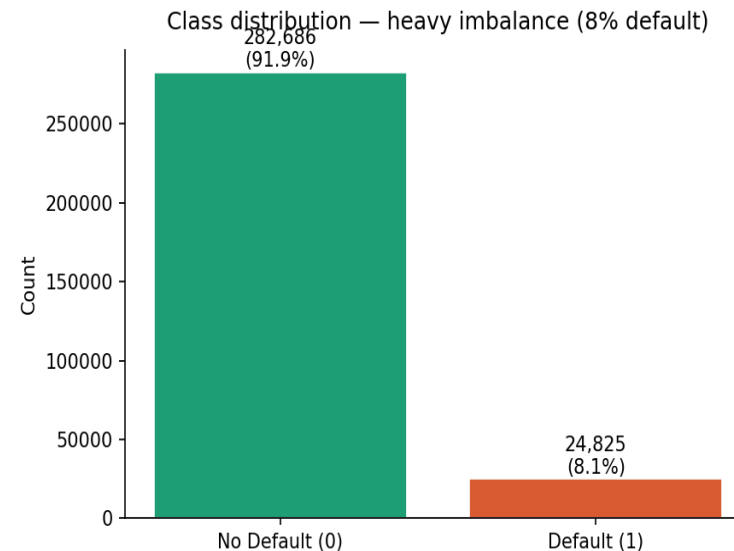
- Credit-to-Income Ratio
- Annuity-to-Income Ratio
- Debt Service Ratio
- Credit Stress Index
- External Source Aggregates

Class Imbalance Strategy

- `scale_pos_weight = 5`
- `Threshold Optimization = 0.30`

Models

- Champion Model: LightGBM
- Challenger Model: XGBoost



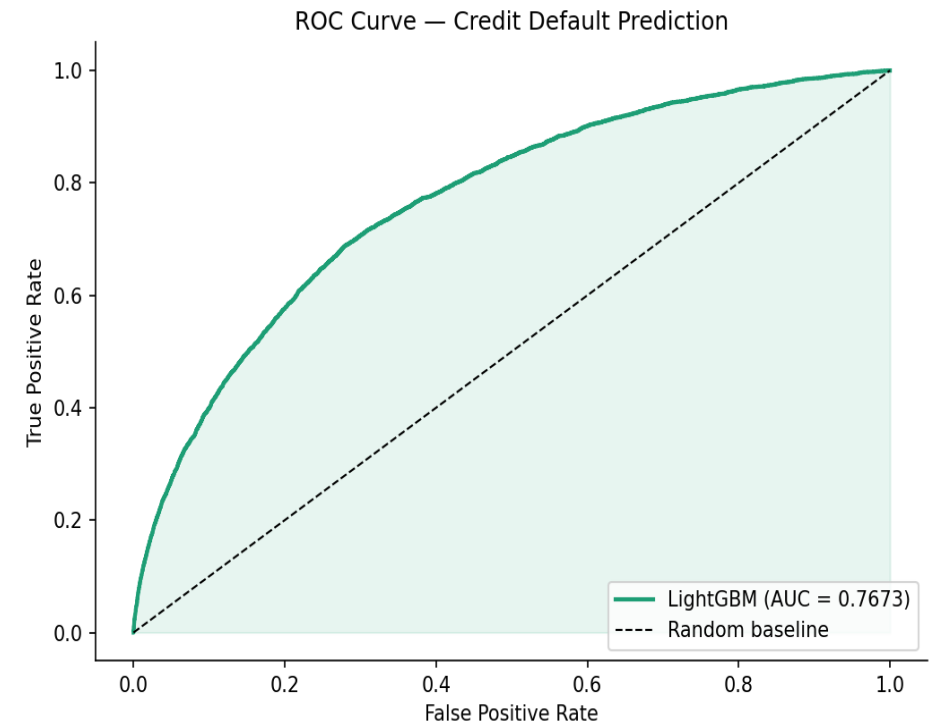
Model Evaluation & Results

Performance Comparison

Metric	LightGBM	XGBoost
ROC-AUC	0.7673	0.7649
PR-AUC	0.2608	0.2578
KS Statistic	0.4089	0.4016

Key Finding

LightGBM demonstrated superior performance across all evaluation metrics and was selected as the production model.



Explainable AI (SHAP)

Global Explainability

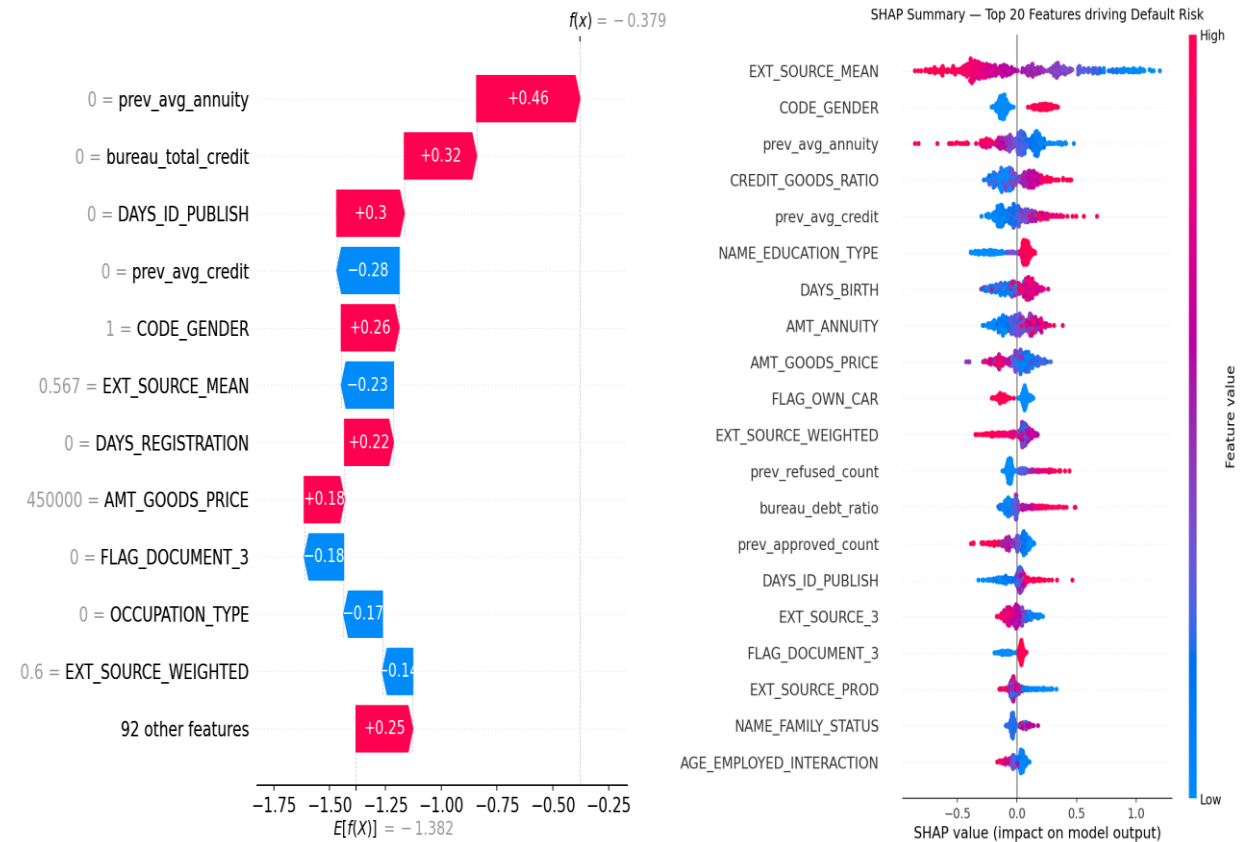
- SHAP Beeswarm plots identify the most influential risk drivers across the portfolio.

Individual Explainability

- SHAP Waterfall plots explain how individual features contribute to a specific prediction.

Business Benefits

- Audit Readiness
- Regulatory Compliance
- Transparent Decision-Making



Risk Prediction Engine

Workflow

- User enters applicant demographics and financial information.
- Engine generates default probability.
- Applicant is assigned a risk band.
- Decision rules provide approval recommendations.

Output

- Risk Score
- Risk Band (Low / Medium / High)
- Approval Recommendation

The screenshot displays the 'Credit Risk Intelligence Platform' interface. It features a dark-themed layout with various input fields and sliders. The input fields include Gender (M), Annual Income (£) (38000), Contract Type (Cash loans), Loan Amount (£) (45000), Income Type (Working), Monthly Annuity (£) (2250), Education (Secondary / secondary special), and Goods Price (£) (45000). Below these are sliders for Age (years) (30), Years Employed (5), and Family Members (2). The 'External Credit Scores' section shows three sliders for EXT_SOURCE_1 (0.50), EXT_SOURCE_2 (0.70), and EXT_SOURCE_3 (0.50). A red bar indicates the 'Assess Risk' process. The output section displays the Risk Score (40.6), Risk Band (Medium), Default Probability (40.6%), and the Approval Recommendation (APPROVE).

Input Field	Value
Gender	M
Annual Income (£)	38000
Contract Type	Cash loans
Loan Amount (£)	45000
Income Type	Working
Monthly Annuity (£)	2250
Education	Secondary / secondary special
Goods Price (£)	45000
Age (years)	30
Years Employed	5
Family Members	2
EXT_SOURCE_1	0.50
EXT_SOURCE_2	0.70
EXT_SOURCE_3	0.50

Output Metric	Value
Risk Score (0-100)	40.6
Risk Band	Medium
Default Probability	40.6%
Approval Recommendation	APPROVE

Conversational Talk-to-Data System

Workflow

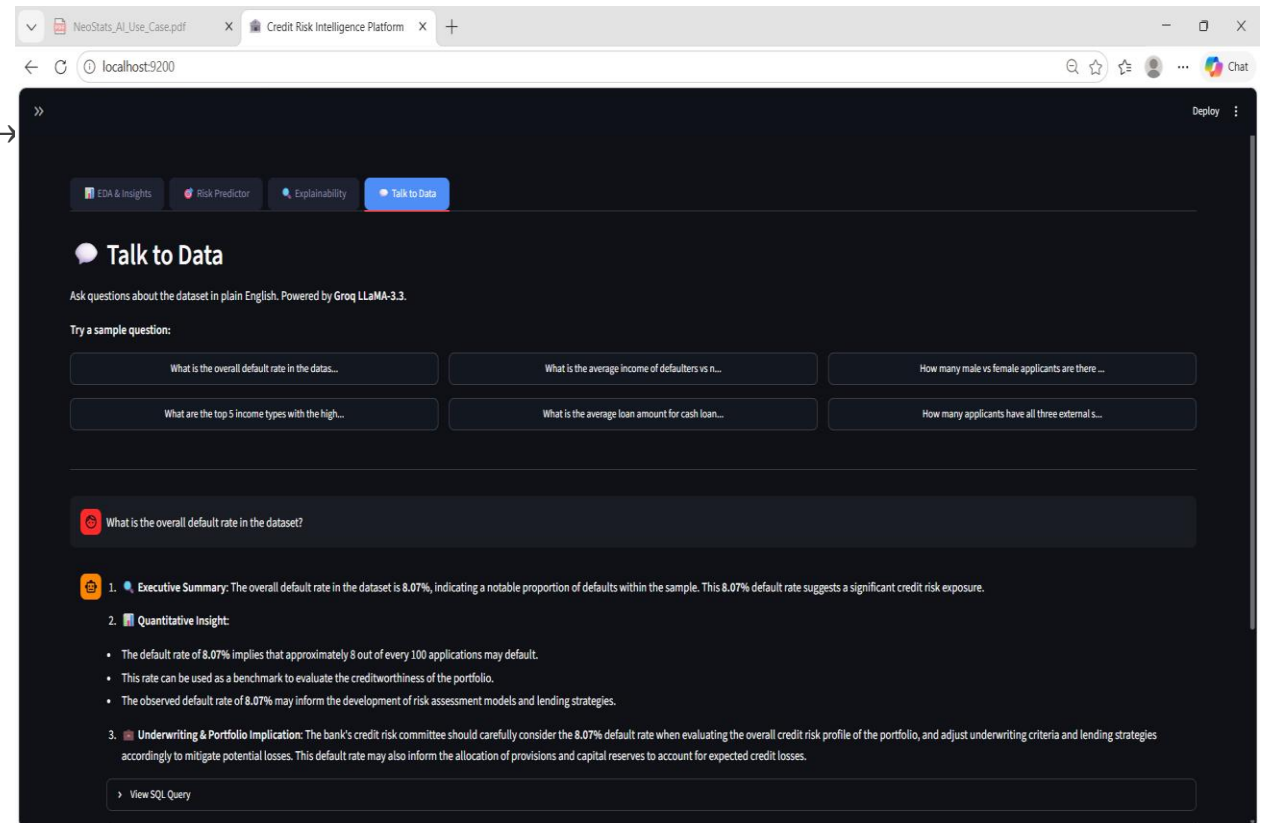
Natural Language → SQL Query → SQLite Execution → Business Insight

Key Features

- NL-to-SQL Translation using Groq LLaMA 3.3
- Schema-Aware Query Generation
- Read-Only SQL Validation
- Hallucination Control Mechanism

Business Benefit

Allows non-technical users to explore datasets without writing SQL.



Business Rules & Underwriting Policy

Rule 1

- High External Credit Score + Low Credit-to-Income Ratio
→ Auto Approve

Rule 2

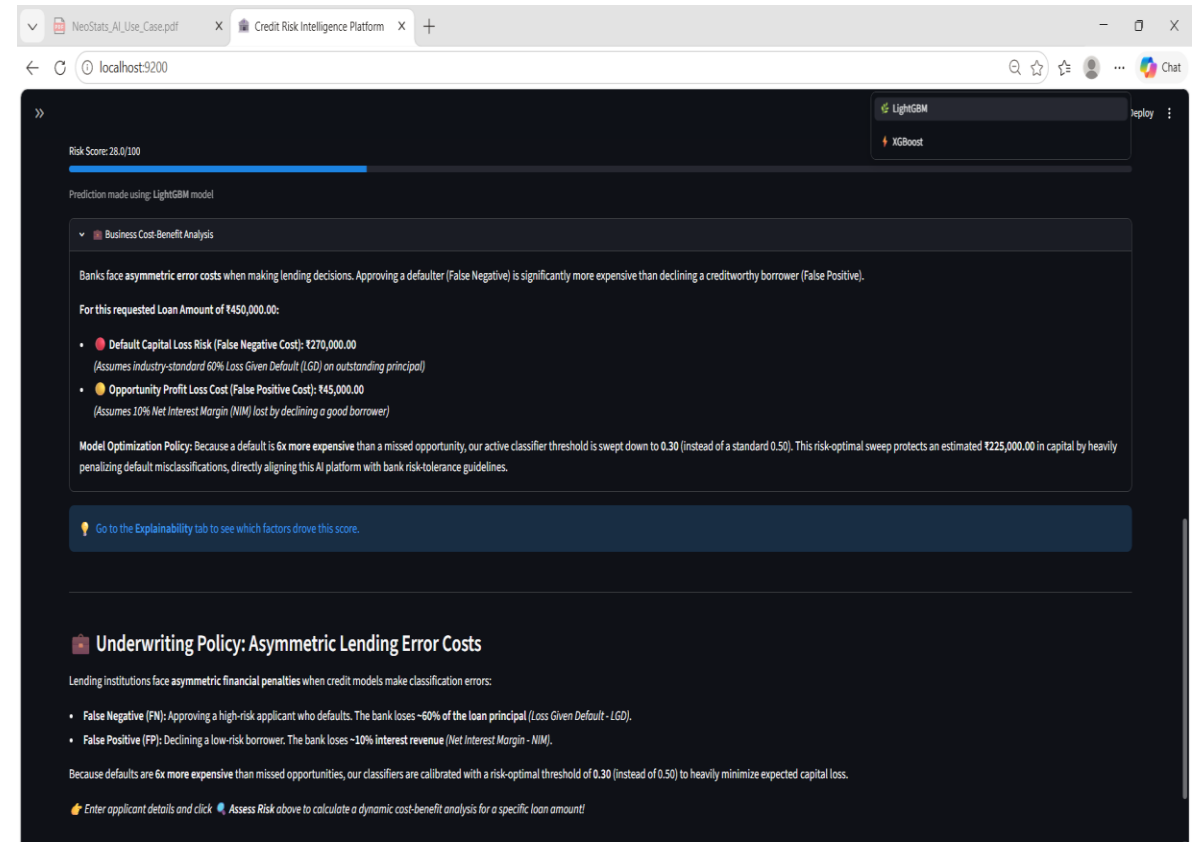
- Debt Service Ratio > 40%
→ Auto Decline

Rule 3

- Age < 30 and Employment Duration < 2 Years
→ Manual Review

Why Rules Matter

Transforms machine learning outputs into actionable credit policy decisions.



Deployment, Impact & Future Work

Deployment

- Dockerized Application
- Docker Compose Orchestration
- SQLite Backend
- Streamlit Frontend

Project Impact

- Automated Credit Risk Assessment
- Explainable AI Decisions
- Conversational Data Analytics
- Improved Decision Support

Future Enhancements

- PostgreSQL Migration
- Cross-Validation Framework
- RAG-Powered Data Assistant
- Real-Time Monitoring

Thank You